

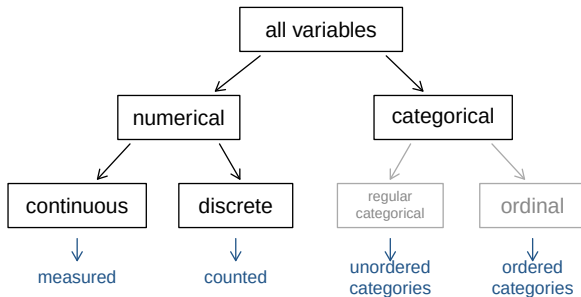
Lecture 2: Data Description

STAT 521

Applied Probability and Statistical Inference

Brenda Betancourt

Types of variables



Variability is a very important concept in statistics. The higher the variability, the harder it is to draw conclusions.

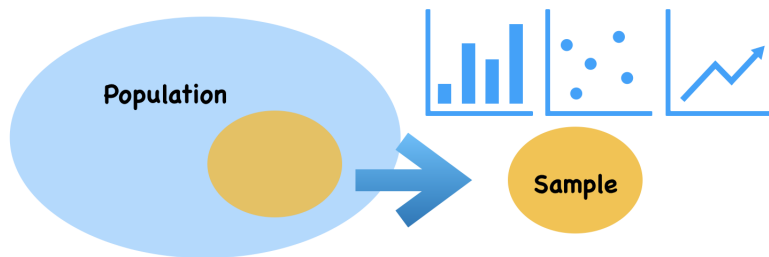
Common sources of variability:

- Variability across measurements e.g. measurement error can cause the recorded length to vary from measurement to measurement.
- Variability across subjects e.g. average pulse rate varies from person to person
- Variability across groups e.g. variability between men and women.

Statistical Studies

Population: All measurements of interest (unknown)

Sample: Subset of population that is obtained to be analyzed (known)



- Sample must reflect the population of interest to infer validly that the results are applicable to a larger group, not just the sample.

Single Variable: Numerical descriptive measures

The most common numerical descriptive measures are:

- Central tendency measures: mean, median, mode
- Variability measures: variance, standard deviation, range, IQR

population $\longrightarrow$ *parameters*

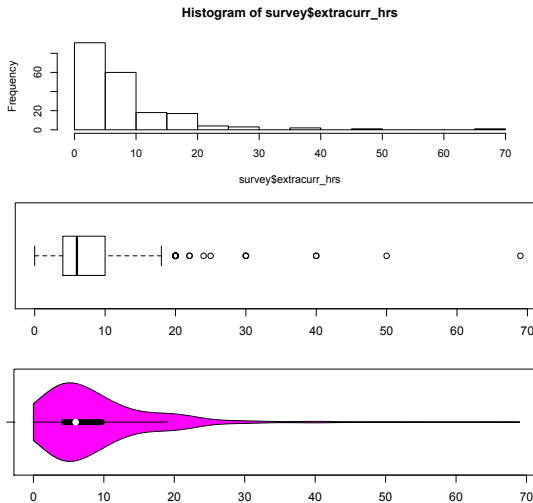
sample $\longrightarrow$ *statistics*

	mean	variance	SD
sample	$\bar{x}$	s^2	s
population	μ	σ^2	σ

Graphical Description of Numeric Variables

- *Histograms*: The heights of the bars above the bins represent the frequency or relative frequency of the various bins.
- *Box plots*: They identify particular percentiles of a distribution and are useful in detecting outlying observations and spread in the distribution.
- *Density plots*: They represent smoothed histograms describing the proportion of measurements within some distance of each point on the continuum.
- *Violin plots*: Similar to box-plot but allows to see better the shape of the distribution.

Univariate Plots - Numerical



Mean

- The *sample mean*, denoted as $\bar{x}$, can be calculated as

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n},$$

where $x_1, x_2, \cdots, x_n$ represent the n observed values.

- The *population mean* is also computed the same way but is denoted as μ . It is often not possible to calculate μ since population data is rarely available.
- The sample mean is a *sample statistic*, or a *point estimate* of the population mean. This estimate may not be perfect, but if the sample is good (representative of the population) it is usually a good guess.

Median

- The *median* is the value that splits the data in half when ordered in ascending order.

$$0, 1, 2, 3, 4$$

- If there are an even number of observations, then the median is the average of the two values in the middle.

$$0, 1, \underline{2}, \underline{3}, 4, 5 \rightarrow \frac{2 + 3}{2} = 2.5$$

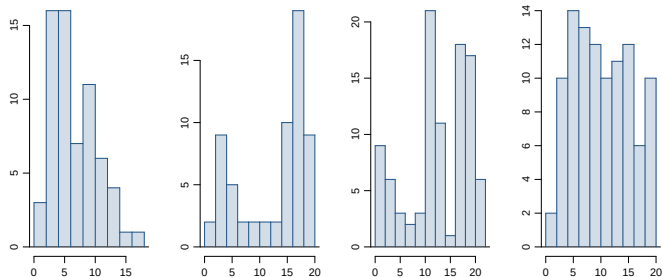
- Since the median is the midpoint of the data, 50% of the values are below it. Hence, it is also the *50th percentile*.

Shape of a distribution: modality

The *mode* is defined as the most frequent observation in the data set.

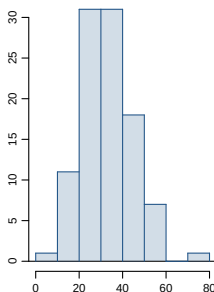
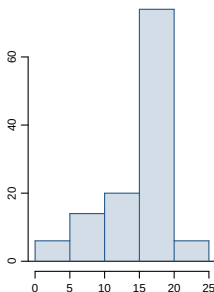
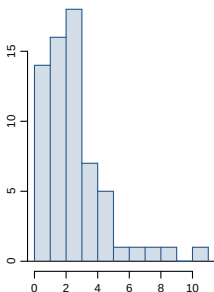
65, 70, 72, 70, 88, 90, 70, 85

Does the histogram have a single prominent peak (*unimodal*), several prominent peaks (*bimodal/multimodal*), or no apparent peaks (*uniform*)?



Shape of a distribution: skewness

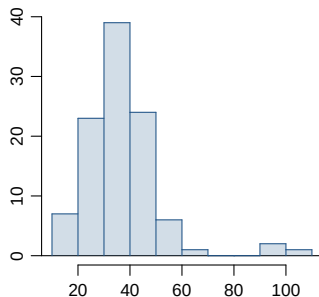
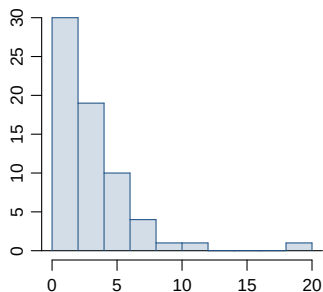
Is the histogram *right skewed*, *left skewed*, or *symmetric*?



Histograms are said to be skewed to the side of the long tail.

Shape of a distribution: unusual observations

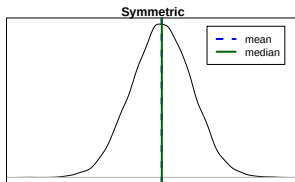
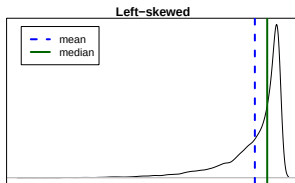
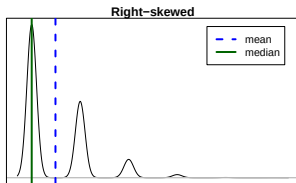
Are there any unusual observations or potential *outliers*? Look for data points that differ markedly from the overall pattern of the data.



They may be unusually large or unusually small compared to the rest of the values.

Mean vs. median

- If the distribution is symmetric, center is the mean
 - Symmetric: mean $\approx$ median
- If the distribution is skewed or has outliers center is the median
 - Right-skewed: mean $>$ median
 - Left-skewed: mean $<$ median

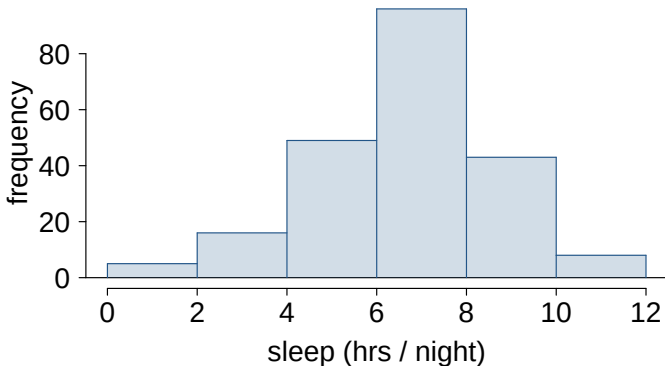


How would you expect each of the following to be distributed?

- Ⓐ Weights of adult males
- Ⓑ Salaries of people from the DMV area
- Ⓒ Exam scores on an easy test
- Ⓓ Birthdays of classmates (day of the month)

Variability in data

How would you describe the amount of variability in the number of hours of sleep students get per night?



Deviation

The distance of an observation from the mean is its *deviation*: $x_i - \bar{x}$.

[illegible]

$$x_1 - \bar{x} = 1 - 6.7 = -5.7$$

$$x_2 - \bar{x} = 1 - 6.7 = -5.7$$

$$x_3 - \bar{x} = 1 - 6.7 = -5.7$$

$$x_4 - \bar{x} = 2 - 6.7 = -4.7$$

•
•
•

$$x_{218} - \bar{x} = 12 - 6.7 = 5.3$$

What is the median and mode?

Variance

Sample *variance*, s^2 , is roughly the average squared deviation from the mean.

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

The variance of amount of sleep students get per night can be calculated as:

$$s^2 = \frac{(-5.7)^2 + (-5.7)^2 + \dots + (5.3)^2}{218 - 1} = \frac{32.6 + 32.6 + \dots + 30.0}{217} = 4.11$$

```
var(d$sleep, na.rm=TRUE)
[1] 4.11
```

Variance (cont.)

Why do we use the squared deviation in the calculation of variance?

- To get rid of negatives so that observations equally distant from the mean are weighed equally.
- To weigh larger deviations more heavily.
- This makes the measure sensitive to outliers and ensures that large differences are emphasized.

Standard deviation

The sample *standard deviation*, s , is the square root of the variance.

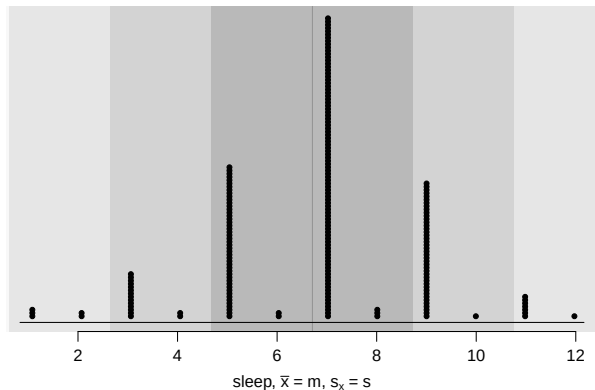
$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

The standard deviation of sleep can be calculated as:

$$s = \sqrt{4.11} = 2.02$$

```
sd(d$sleep, na.rm=TRUE)
[1] 2.02
```

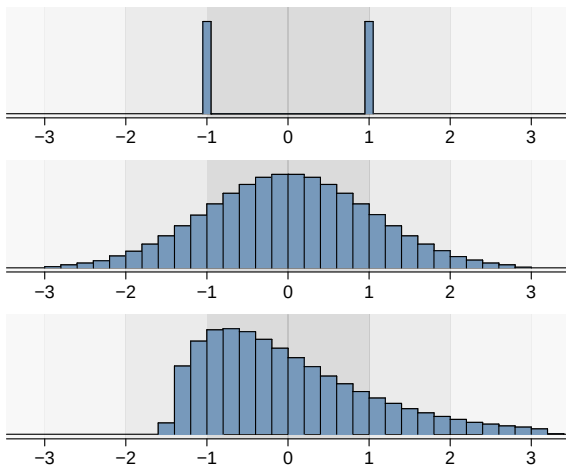
Normal link: Empirical rule for symmetric distributions



- 145 out of 218 students (67%) are within 1 SD of the mean.
- 204 out of 218 students (94%) are within 2 SDs of the mean.
- 217 out of 218 students (100%) are within 3 SDs of the mean.

Describing distributions

When describing distributions make sure to talk about the shape, center, spread, and if any, unusual observations.



Q1, Q3, and IQR

- The 25th percentile is also called the first quartile, *Q1*.
- The 50th percentile is also called the median (2nd quartile), *Q2*.
- The 75th percentile is also called the third quartile, *Q3*.

```
summary(d$study_hours)
```

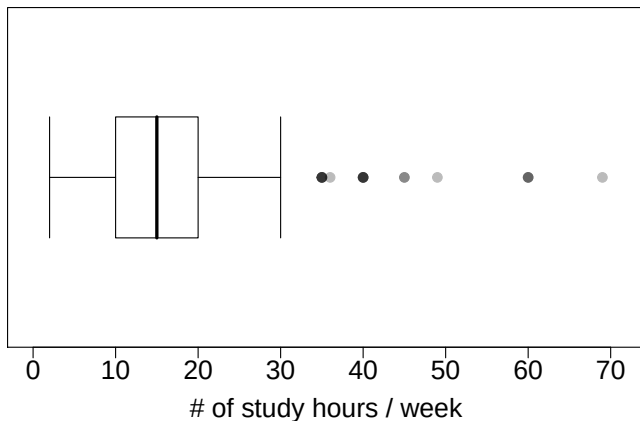
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
2.00	10.00	15.00	17.45	20.00	69.00	20.00

- Between Q1 and Q3 is the middle 50% of the data. The range these data span is called the *interquartile range*, or the *IQR*.

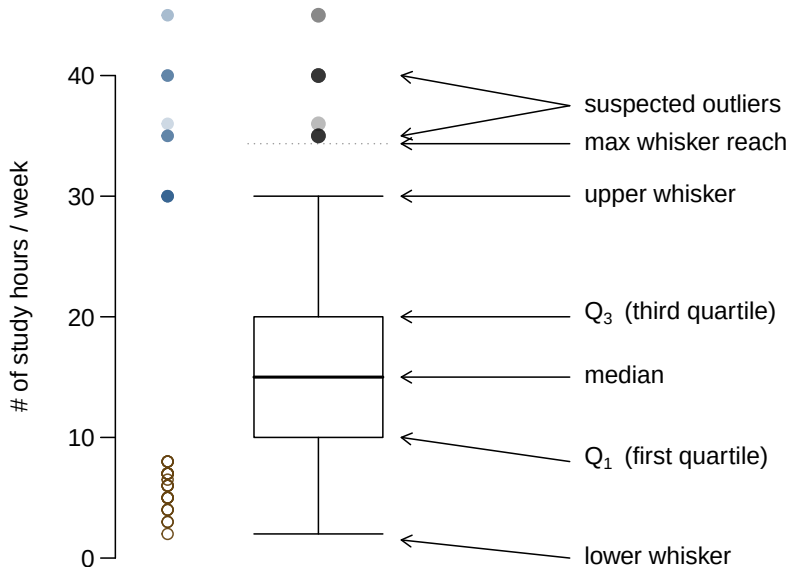
$$IQR = 20 - 10 = 10$$

Box plot

The box in a *box plot* represents the middle 50% of the data, and the thick line in the box is the median.



Anatomy of a box plot



Whiskers and outliers

- *Whiskers* of a box plot can extend up to $1.5 * IQR$ away from the quartiles.

$$\text{max upper whisker reach : } Q3 + 1.5 * IQR = 20 + 1.5 * 10 = 35$$

$$\text{max lower whisker reach : } Q1 - 1.5 * IQR = 10 - 1.5 * 10 = -5$$

- The *whiskers* are calculated as the min/max value between the data points within the max whisker reach limit:

$$\text{upper whisker : } \max(x) \leq Q3 + 1.5 * IQR \rightarrow 30$$

$$\text{lower whisker : } \min(x) \geq Q1 - 1.5 * IQR \rightarrow 2$$

- An *outlier* is defined as an observation beyond the maximum reach of the whiskers.

Outliers (cont.)

Why is it important to look for outliers?

- Identify extreme skewness in the distribution.
- Identify data collection and entry errors.
- Provide insight into interesting features of the data.

Household income

Which of the below is the most reasonable estimate for the median household income? (n = 148)



- (a) \$50K
- (b) \$150K *\$150K*
- (c) \$300K
- (d) \$400K
- (e) \$500K

Extreme observations

How would sample statistics such as mean, median, SD, and IQR of household income be affected if the largest value was replaced with \$10 million? What if the smallest value was replaced with \$10 million?



Robust statistics



scenario	robust		not robust	
	median	IQR	$\bar{x}$	s
original data	165K	150K	211K	180K
move largest to \$10 million	165K	150K	398K	1,422K
move smallest to \$10 million	190K	163K	4,186K	1,424K

Robust statistics

Median and IQR are more robust to skewness and outliers compared to the mean and SD. Therefore,

- For skewed distributions it is more appropriate to use median and IQR to describe the center and spread
- For symmetric distributions it is more appropriate to use the mean and SD to describe the center and spread

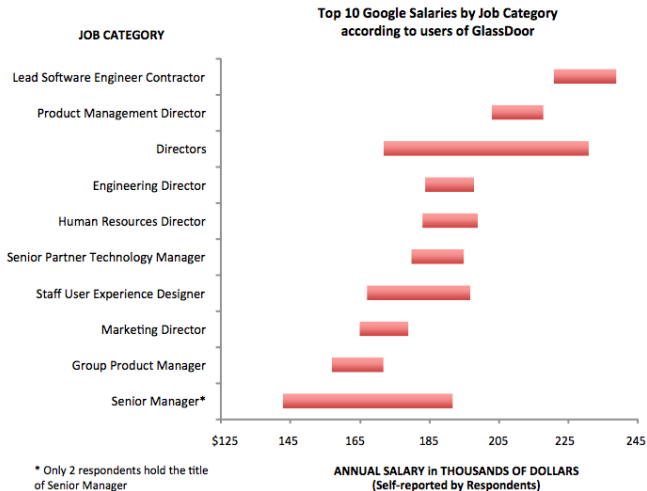
If you would like to estimate the typical household income for a GMU student, would you be more interested in the mean or median income?

More on Graphical Description

Depending on the type of measurement, common plots are pie charts, bar charts, histograms, box plots, and density plots.

- *Pie charts:* Used to describe any variable type.
 - Continuous numeric variables must be collapsed into bins or buckets.
 - The size of the sectors of the pie represent the relative frequency of each category.
- *Bar charts:* Nominal or ordinal data.
 - The variable levels are arrayed on the bottom (or left side) of the plot.
 - Bars above (or beside) the levels represent the range of values taken by the response variable or the frequency or relative frequency of the number of observations belonging to the various categories.

Example: Barchart

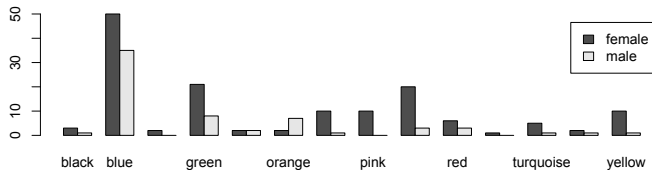
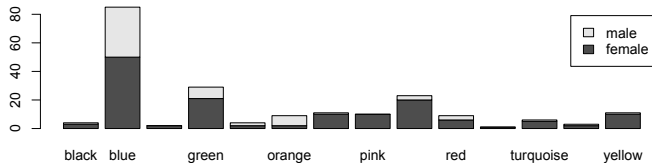


Graphical description: Bivariate Plots

- Categorical vs Categorical
 - Stacked or grouped bar plot
- Numerical vs Categorical
 - Side by side boxplot or violin plot
- Numerical vs Numerical
 - Scatter plot, line plot

Bivariate Plots - Bar plots

Favorite color by gender:



Bivariate Plots - Scatter plot

The *scatterplot* below shows the relationship between HS graduate rate in all 50 states and the District of Columbia vs the % of residents who live below the poverty line ($< \$22,350$ for a family of 4).

